Stochastic ordering in multivariate extremes

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Joint work with Michela Corradini

Outline

1. **Recalling insights from** Multivariate extremes
2. **Background on** Stochastic orderings
3. **Results**

https://link.springer.com/article/10.1007/s10687-024-00486-0

Multivariate extremes

Structure of multivariate max-stable distributions G

Representations of G up to marginal standardisation (unit Fréchet margins):

- by a homogeneous exponent measure Λ: [Resnick 1987]

$$G(\mathbf{x}) = \exp\big\{-\Lambda([\mathbf{0},\mathbf{x}]^c)\big\}$$

- by a spectral representation $(\Omega, \mathcal{A}, \nu, \mathbf{f})$: [de Haan 1984]

$$G(\mathbf{x}) = \exp\left\{-\int_\Omega \max_{i=1,\dots,d} \frac{f_i(\omega)}{x_i}\nu(d\omega)\right\}$$

  special cases:
  angular measure: $\Omega = \triangle_d = \{\omega \in [0,\infty)^d : \|\omega\|_1 = 1\}$, $\mathbf{f}$ coordinate maps,
  generator: $(\Omega, \mathcal{A}, \nu)$ probability space, $\mathbf{f} = \mathbf{Z}$ [Falk 2019]

- by a homog. max-completely alternating stable tail dependence function ℓ: [Ressel 2013]

$$G(\mathbf{x}) = \exp\big\{-\ell\big(1/x_1, \dots, 1/x_d\big)\big\}$$

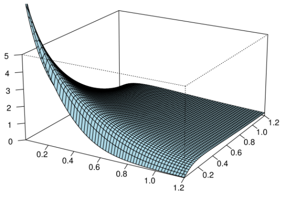
  (restricted to $\triangle_d$ called Pickands dependence function)

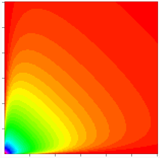
- by a max-zonoid $K = \{\mathbf{k} : \langle \mathbf{k}, \cdot \rangle \leqslant \ell\}$, here $\ell = \sup\{\langle \cdot, \mathbf{k}\rangle : \mathbf{k} \in K\}$ [Molchanov 2008]

- respective copulas are extreme value copulas [Gudendorf/Segers 2010]

Properties of the exponent measure Λ

- finite away from the origin $0 \in \mathbb{R}^d$
- homogeneous: $\Lambda(tA) = t^{-\alpha}\Lambda(A)$
- explodes at 0, but $\Lambda(\{0\}) = 0$
- can be seen as Lévy-measure wrt max:
  $X \sim G_\Lambda$ is **max-infinite divisible**,
  $X \sim \max_{i=1}^{\infty} \eta_i$, where $\{\eta_i\} \sim$ PPP(Λ).





Examples of parametric families

(closed under taking marginal distributions)

- Max-stable Dirichlet model

$$\mathrm{MaxDir}(\alpha_1, \alpha_2, \ldots, \alpha_d), \quad (\alpha_1, \ldots, \alpha_d) \in (0, \infty)^d$$

- Hüsler-Reiss distribution

$$\mathrm{HR}(\{\gamma_{ij}\}_{i,j=1,\ldots,d}), \quad \{\gamma_{ij}\}_{i,j=1,\ldots,d} \text{ conditionally negative definite}$$

- Choquet/Tawn-Molchanov model

$$\mathrm{Choquet}(\{\theta_A\}_{\emptyset\neq A\subset\{1,\ldots,d\}}), \quad \{\theta_A\}_{\emptyset\neq A\subset\{1,\ldots,d\}} \text{ extremal coefficients,}$$
$$\mathrm{Choquet}(\{\chi_A\}_{\emptyset\neq A\subset\{1,\ldots,d\}}) \quad \{\chi_A\}_{\emptyset\neq A\subset\{1,\ldots,d\}} \text{ tail dependence coefficients}$$

Here, $\chi = T(\theta)$ and $\theta = T(\chi)$, where

$$T : \mathbb{R}^{\mathcal{P}_d\setminus\{\emptyset\}} \to \mathbb{R}^{\mathcal{P}_d\setminus\{\emptyset\}}, \quad \{x_A\} \mapsto \{y_A\} \text{ with } y_A = \sum_{I\subset A, I\neq\emptyset} (-1)^{|I|+1} x_I.$$

Max-stable Dirichlet model

(i) New? **Gamma generator**:

$$\alpha^{-1}\mathbf{\Gamma} = (\Gamma_1/\alpha_1, \Gamma_2/\alpha_2, \dots, \Gamma_d/\alpha_d)^\top,$$

where $\mathbf{\Gamma} = (\Gamma_1, \dots, \Gamma_d)^\top$ consists of independent Gamma distributed variables $\Gamma_i \sim \Gamma(\alpha_i)$, $\alpha_i > 0$, $i = 1, \dots, d$.

(ii) **Dirichlet generator**:

$$(\alpha^{-1}\|\alpha\|_1)\mathbf{D} = (\alpha_1 + \cdots + \alpha_d) \cdot (D_1/\alpha_1, D_2/\alpha_2, \dots, D_d/\alpha_d)^\top,$$

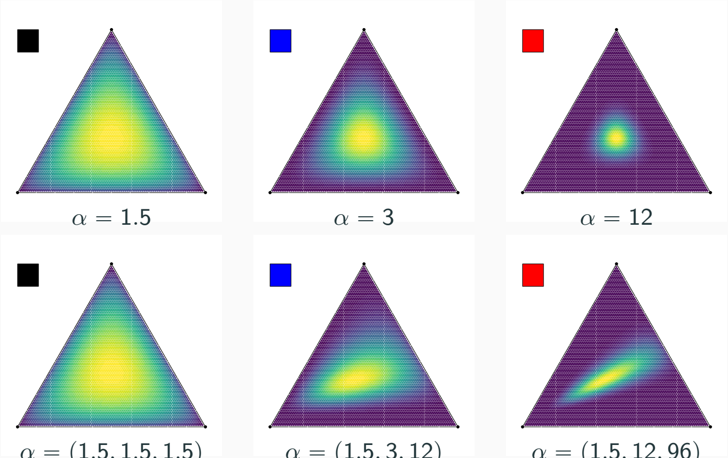
where $\mathbf{D}$ follows a Dirichlet distribution $\mathrm{Dir}(\alpha_1, \dots, \alpha_d)$ on the unit simplex $\triangle_d$

(iii) **Angular measure density** on $\triangle_d$:

$$h(\omega_1, \dots, \omega_d) = \frac{\Gamma(\|\alpha\|_1 + 1)}{\|\alpha\omega\|_1} \prod_{i=1}^{d} \frac{\alpha_i^{\alpha_i} \omega_i^{\alpha_i - 1}}{\Gamma(\alpha_i)(\|\alpha\omega\|_1)^{\alpha_i}}, \quad (\omega_1, \dots, \omega_d)^\top \in \triangle_d.$$

Fun fact: **Gamma generator** helps to demonstrate that the model is marginally closed; previously only known from lengthy density calculations in [Ballani/Schlather 2011]

Example (Dirichlet model) - Angular densities (heat maps)



$\alpha = 1.5$ $\quad$ $\alpha = 3$ $\quad$ $\alpha = 12$

$\alpha = (1.5, 1.5, 1.5)$ $\quad$ $\alpha = (1.5, 3, 12)$ $\quad$ $\alpha = (1.5, 12, 96)$

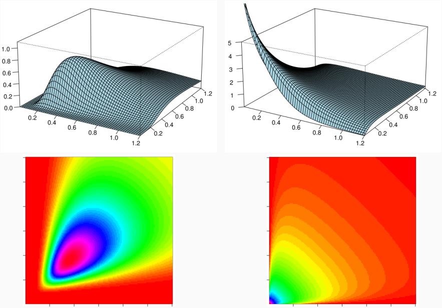
Hüsler-Reiss distribution

Generator:
$\mathbf{Z} = (Z_1, \ldots, Z_d)^\top$ with

$$Z_i = \exp\left(W_i - \frac{1}{2}\mathrm{Var}(W_i)\right),$$

where
$\mathbf{W} = (W_1, \ldots, W_d)^\top$ is zero mean Gaussian with variogram (incremental variance)

$$\mathbb{E}(W_i - W_j)^2 = \gamma_{ij}.$$



Density and exponent measure of a bivariate HR model

Choquet/Tawn-Molchanov model

Three different parametrizations for the Choquet model:

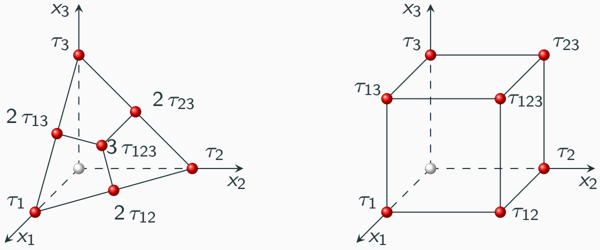
(i) by the $2^d - 1$ extremal coefficients $\theta(A)$, $A \in \mathcal{P}_d$, $A \neq \emptyset$,

(ii) by the $2^d - 1$ tail dependence coefficients $\chi(A)$, $A \in \mathcal{P}_d$, $A \neq \emptyset$,

(iii) by the $2^d - 1$ mass coefficients $\tau(A)$, $A \in \mathcal{P}_d$, $A \neq \emptyset$.

Only (i) and (ii) do not depend on the dimension of the model.

Support sets for the angular/spectral measure:



Stochastic orderings

Stochastic orderings

Univariate: Usual stochastic order

$$F \leqslant_{\mathrm{st}} G \quad \text{if} \quad F \geqslant G$$

Draws from the dominating measure are more likely to be large.

Multivariate: plenty! - focus here on

Upper orthants $U_{\mathbf{a}} = \{\mathbf{x} : x_i > a_i,\ i = 1, \ldots, d\}$

$$F \leqslant_{\mathrm{uo}} G \quad \text{if} \quad F(U) \leqslant G(U) \text{ for all upper orthants } U$$

Lower orthants $L_{\mathbf{a}} = \{\mathbf{x} : x_i \leqslant a_i,\ i = 1, \ldots, d\}$

$$F \leqslant_{\mathrm{lo}} G \quad \text{if} \quad F(L) \geqslant G(L) \text{ for all lower orthants } L$$

Systematic background and **closure properties**

(indep/identical concaternation, margins, limits, mixtures, increasing transformations):

[Müller/Stoyan (2002) – Comparison Methods for Stochastic Models and Risks]
[Shaked/Shanthikumar (2007) – Stochastic Orders]

Stochastic orderings

For non-negative $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^d$

$$\mathbf{X} \leqslant_{\mathrm{uo}} \mathbf{Y} \quad \Longleftrightarrow \quad \min_{i=1,\dots,d}(a_i X_i) \leqslant_{\mathrm{st}} \min_{i=1,\dots,d}(a_i Y_i), \quad \mathbf{a} \in (0,\infty)^d;$$

$$\mathbf{X} \leqslant_{\mathrm{lo}} \mathbf{Y} \quad \Longleftrightarrow \quad \max_{i=1,\dots,d}(a_i X_i) \leqslant_{\mathrm{st}} \max_{i=1,\dots,d}(a_i Y_i), \quad \mathbf{a} \in (0,\infty)^d.$$

Positive quadrant order (PQD)

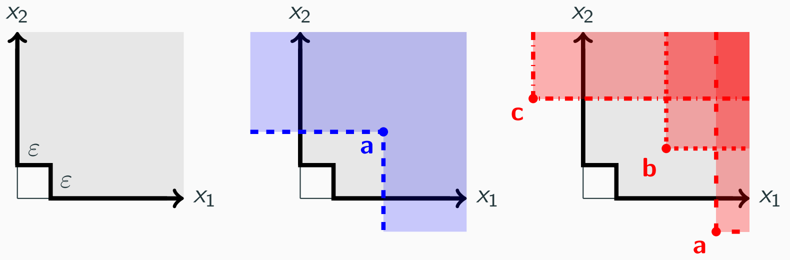
$$\mathbf{X} \leqslant_{\mathrm{PQD}} \mathbf{Y} \quad \text{if} \quad \underbrace{\mathbf{X} \leqslant_{\mathrm{uo}} \mathbf{Y} \quad \text{and} \quad \mathbf{X} \geqslant_{\mathrm{lo}} \mathbf{Y}}_{\textit{subtle}!}$$

Extension to exponent measures Λ, $\widetilde{\Lambda}$

How to make sense of

$$\boxed{\Lambda \leqslant_{\mathrm{lo}} \widetilde{\Lambda}, \quad \Lambda \leqslant_{\mathrm{uo}} \widetilde{\Lambda}, \quad \Lambda \leqslant_{\mathrm{PQD}} \widetilde{\Lambda}} \qquad ?$$

Change test sets to survival type version.
Only test where exponent measures are finite.
Don't forget to include the axes!



Λ, $\widetilde{\Lambda}$ finite for grey areas — Admissable complement of $L_{\mathbf{a}}$ — Admissable $U_{\mathbf{a}}$, $U_{\mathbf{b}}$, $U_{\mathbf{c}}$

Results

Fundamental results

Setup

G, $\widetilde{G}$	–	multivariate max-stable distributions (same margins)
Λ, $\widetilde{\Lambda}$	–	their exponent measures
Z, $\widetilde{Z}$	–	their generators (not unique!)
ℓ, $\widetilde{\ell}$	–	their stable tail dependence functions
K, $\widetilde{K}$	–	their max-zonoids

Easy/well-known:

$$G \leqslant_{\mathrm{lo}} \widetilde{G} \quad \iff \quad \Lambda \leqslant_{\mathrm{lo}} \widetilde{\Lambda} \quad \iff \quad \ell \leqslant \widetilde{\ell} \quad \iff \quad K \subset \widetilde{K}$$

$$\iff \quad \mathbb{E}\Big\{ \max_{i=1,\dots,d} (a_i Z_i) \Big\} \leqslant \mathbb{E}\Big\{ \max_{i=1,\dots,d} (a_i \widetilde{Z}_i) \Big\} \text{ for all } \mathbf{a} \in (0,\infty)^d$$

New/not straightforward:

$$G \leqslant_{\mathrm{uo}} \widetilde{G} \quad \iff \quad \Lambda \leqslant_{\mathrm{uo}} \widetilde{\Lambda} \qquad \text{[sketch of proof]}$$

$$\iff \mathbb{E}\Big\{ \min_{i \in A} (a_i Z_i) \Big\} \leqslant \mathbb{E}\Big\{ \min_{i \in A} (a_i \widetilde{Z}_i) \Big\} \text{ for all } \mathbf{a} \in (0,\infty)^d,\ \emptyset \neq A \subset \{1,\dots,d\}$$

Consequences

- $G \leqslant_{\mathrm{PQD}} \widetilde{G} \quad \Longleftrightarrow \quad \Lambda \leqslant_{\mathrm{PQD}} \widetilde{\Lambda}$
- For any (simple) max-stable G

$$G_{\mathrm{indep}} \quad \leqslant_{\mathrm{PQD}} \quad G \quad \leqslant_{\mathrm{PQD}} \quad G_{\mathrm{dep}}.$$

Actually, slightly more:

- If G^* is the associated Choquet model to G, then

$$G_{\mathrm{indep}} \quad \leqslant_{\mathrm{PQD}} \quad G^* \quad \leqslant_{\mathrm{PQD}} \quad G \quad \leqslant_{\mathrm{PQD}} \quad G_{\mathrm{dep}}.$$

[explain terms]

Results for parametric models

Dirichlet family:

Theorem

If $\alpha_i \leqslant \beta_i$, $i = 1, \ldots, d$, then

$$\mathrm{MaxDir}(\alpha_1, \alpha_2, \ldots, \alpha_d) \leqslant_{\mathrm{PQD}} \mathrm{MaxDir}(\beta_1, \beta_2, \ldots, \beta_d).$$

[Proof based on convolution stability of Gamma generator.]

Hüsler-Reiss family:

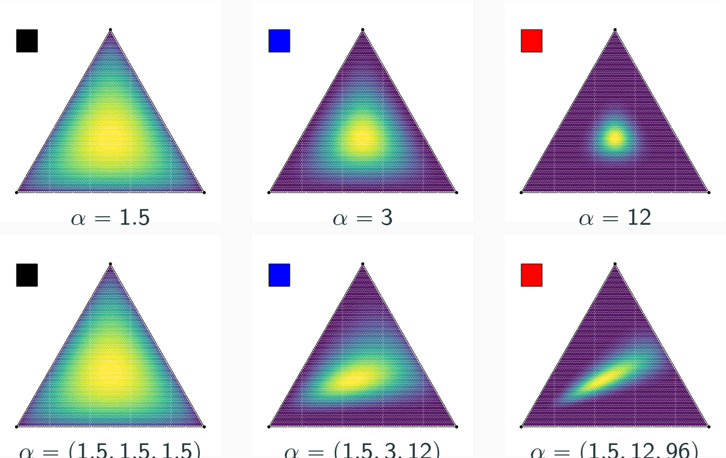
Theorem

If $\gamma_{i,j} \leqslant \tilde{\gamma}_{i,j}$ for all $i, j \in \{1, \ldots, d\}$, then

$$\mathrm{HR}(\gamma) \geqslant_{\mathrm{PQD}} \mathrm{HR}(\tilde{\gamma}).$$

[Proof based on obtaining HR model as triangular array limit of Gaussians with correlations $\exp\{-\gamma_{ij}/c_n\}$ and Slepian's normal comparison lemma.]

Example (Dirichlet model) - Angular densities (heat maps)



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- corresponding max-stable distributions PQD-increasing from left to right

Is everything in order?

Choquet/TM model yields (counter-)examples for various situations:

Lemma (Stochastic ordering for Choquet/TM model)

$$\theta \leqslant \widetilde{\theta} \qquad \Longleftrightarrow \qquad \mathrm{Choquet}_{\mathrm{EC}}(\theta) \leqslant_{\mathrm{lo}} \mathrm{Choquet}_{\mathrm{EC}}(\widetilde{\theta});$$
$$\chi \leqslant \widetilde{\chi} \qquad \Longleftrightarrow \qquad \mathrm{Choquet}_{\mathrm{TD}}(\chi) \leqslant_{\mathrm{uo}} \mathrm{Choquet}_{\mathrm{TD}}(\widetilde{\chi}).$$

One can construct Choquet models A, B, C, D, such that:

- $B \leqslant_{\mathrm{uo}} D$, no order between B and D according to LO;
- $C \leqslant_{\mathrm{lo}} B$, no order between B and C according to UO;
- $A \leqslant_{\mathrm{PQD}} B$;
- $A \leqslant_{\mathrm{uo}} C$ and $A \leqslant_{\mathrm{lo}} C$.

Conclusions

Thank you!

Context:

- multivariate max-stable distributions arise in (non-central) limit theorems and appear in the analysis of dependencies among risks
- knowledge about stochastic orderings among them so far limited mostly to lower orthants / maxima

New insights:

- more understanding around stochastic orderings with respect to upper orthants / minima / PQD
- fundamental relation between distribution G and exponent measure Λ (LO/UO/PQD orders equivalent)
- further relations between popular parametric models (Dirichlet/Hüsler-Reiss)
- warning: stochastic orders are not valid already on generator level
- Choquet models reveal that possibilities are richer

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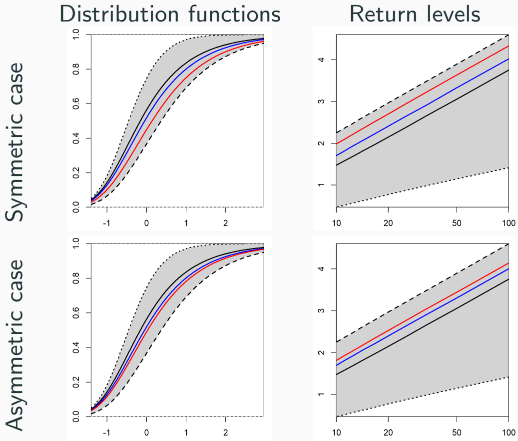
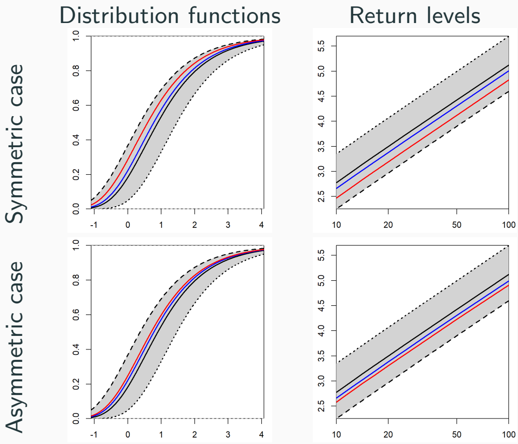
Appendix

Usual stochastic orders for resulting minima



Distribution functions (left) and return levels (right) for return periods 10 to 100 (on logarithmic scale) of $\min(X_1, X_2, X_3)$, where $(X_1, X_2, X_3) \sim \mathrm{MaxDir}(\alpha_1, \alpha_2, \alpha_3)$ on standard Gumbel scale with $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ as chosen in previous figure. The grey areas represent the range between the fully dependent (dashed line) and fully independent (dotted line) cases.

Usual stochastic orders for resulting maxima



Distribution functions (left) and return levels (right) for return periods 10 to 100 (on logarithmic scale) of max(X_1, X_2, X_3), where $(X_1, X_2, X_3) \sim \mathrm{MaxDir}(\alpha_1, \alpha_2, \alpha_3)$ on standard Gumbel scale with $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ as chosen in previous figure. The grey areas represent the range between the fully dependent (dashed line) and fully independent (dotted line) cases.